

AidBridge — ERD v3.0

Engine: PostgreSQL 15+ · Extension: uuid-oss

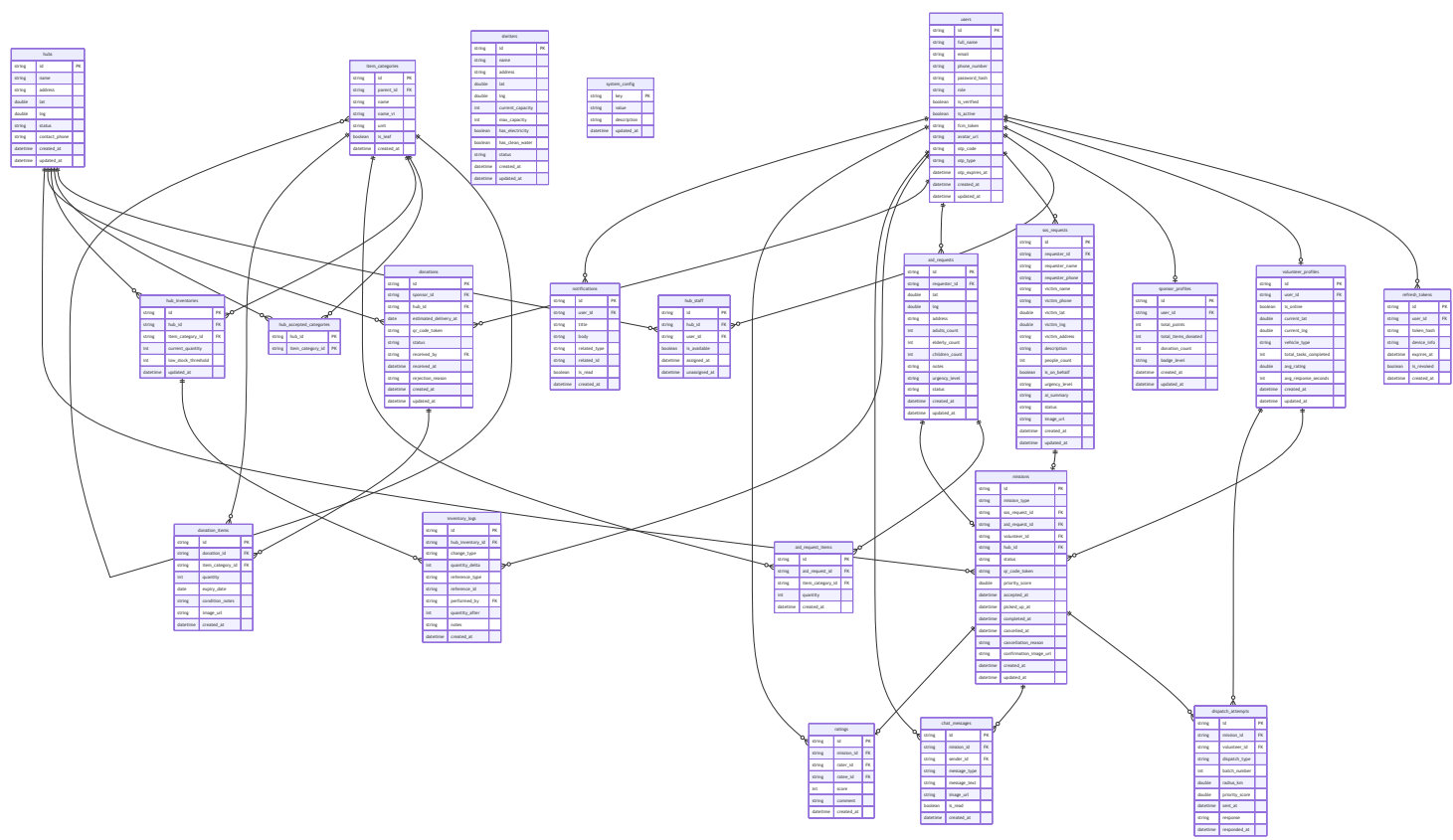
22 bảng · 10 ENUM · ~35 index

Nguyên tắc: lat/lng DECIMAL(9,6) (không PostGIS), image_url VARCHAR trực tiếp trong bảng, OTP gộp vào users .

1. Schema Groups

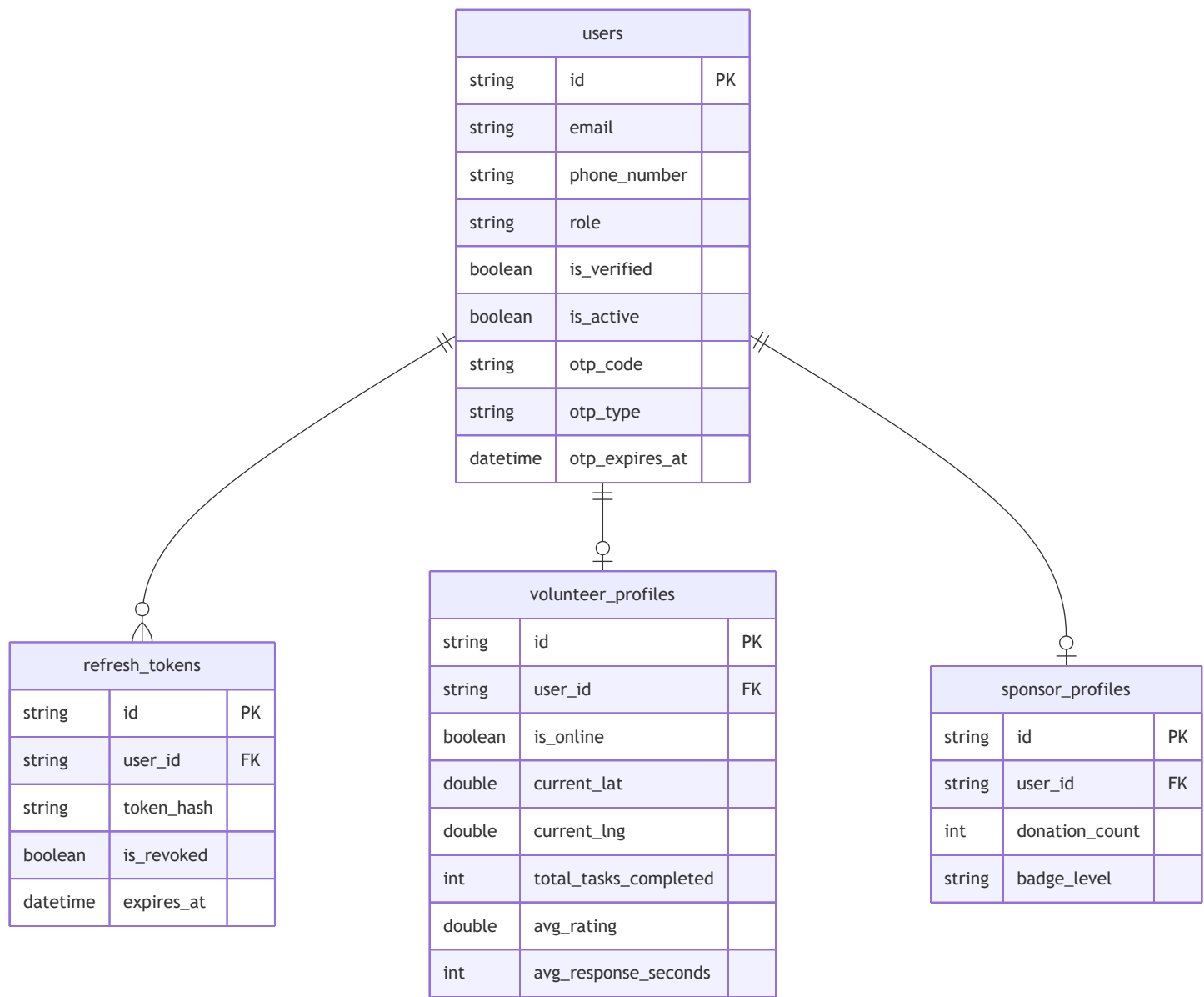
Group	Bảng
Auth	users , refresh_tokens
Profiles	volunteer_profiles , sponsor_profiles
Infrastructure	hubs , hub_staff , shelters , system_config
Catalog & Inventory	item_categories , hub_accepted_categories , hub_inventories , inventory_logs
Requests	sos_requests , aid_requests , aid_request_items
Donations	donations , donation_items
Missions	missions , dispatch_attempts
Communication	chat_messages , ratings , notifications

2. Full ERD

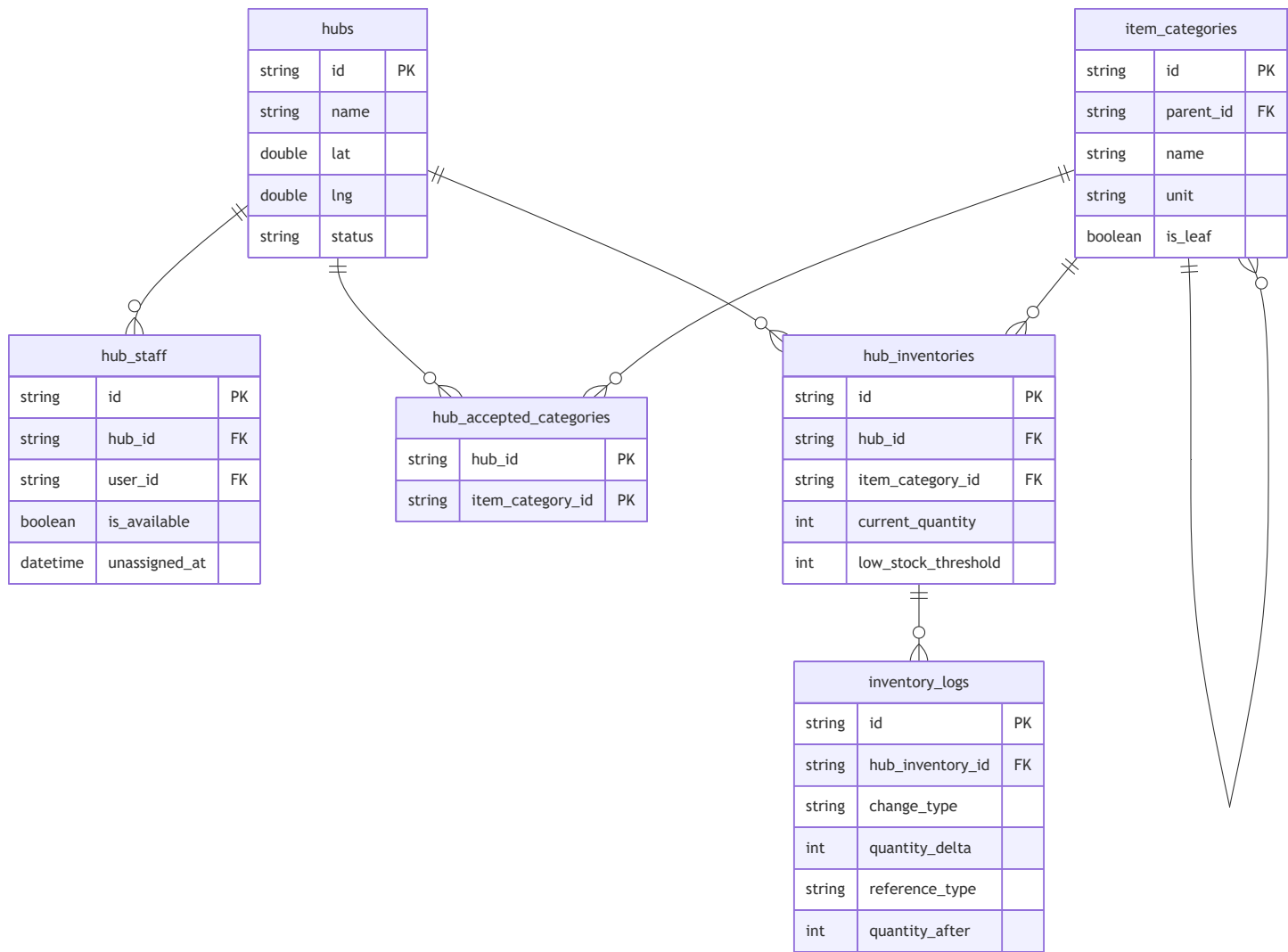


3. Sub-Diagrams theo Domain

A — Auth & Profiles



B — Infrastructure & Inventory



C — Requests & Donations

sos_requests		
string	id	PK
string	requester_id	FK
double	victim_lat	
double	victim_lng	
string	urgency_level	
string	status	
string	image_url	

aid_requests		
string	id	PK
string	requester_id	FK
double	lat	
double	lng	
string	urgency_level	
string	status	

donations		
string	id	PK
string	sponsor_id	FK
string	hub_id	FK
string	qr_code_token	
string	status	

aid_request_items		
string	id	PK
string	aid_request_id	FK
string	item_category_id	FK
int	quantity	

donation_items		
string	id	PK
string	donation_id	FK
string	item_category_id	FK
int	quantity	
string	image_url	

D — Missions & Dispatch

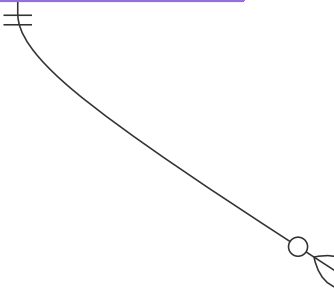
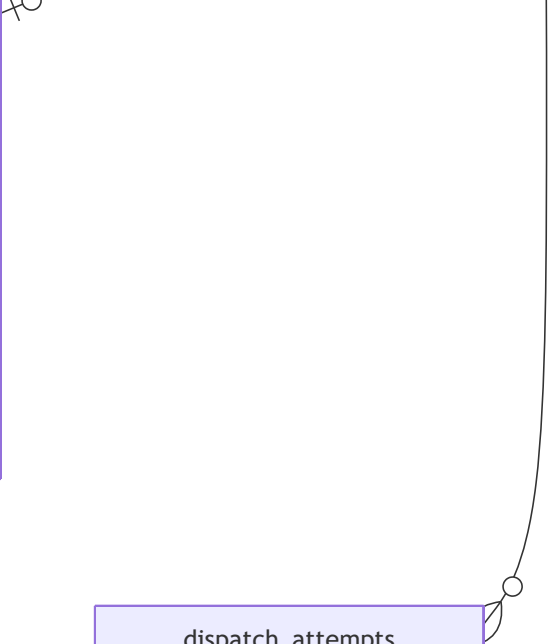
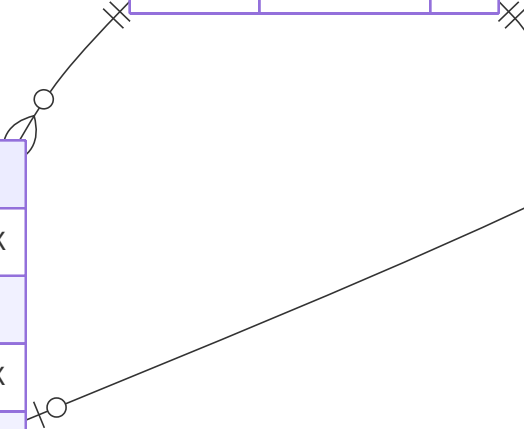
sos_requests		
string	id	PK
string	status	

volunteer_profiles		
string	id	PK
boolean	is_online	
double	current_lat	
double	current_lng	
double	avg_rating	

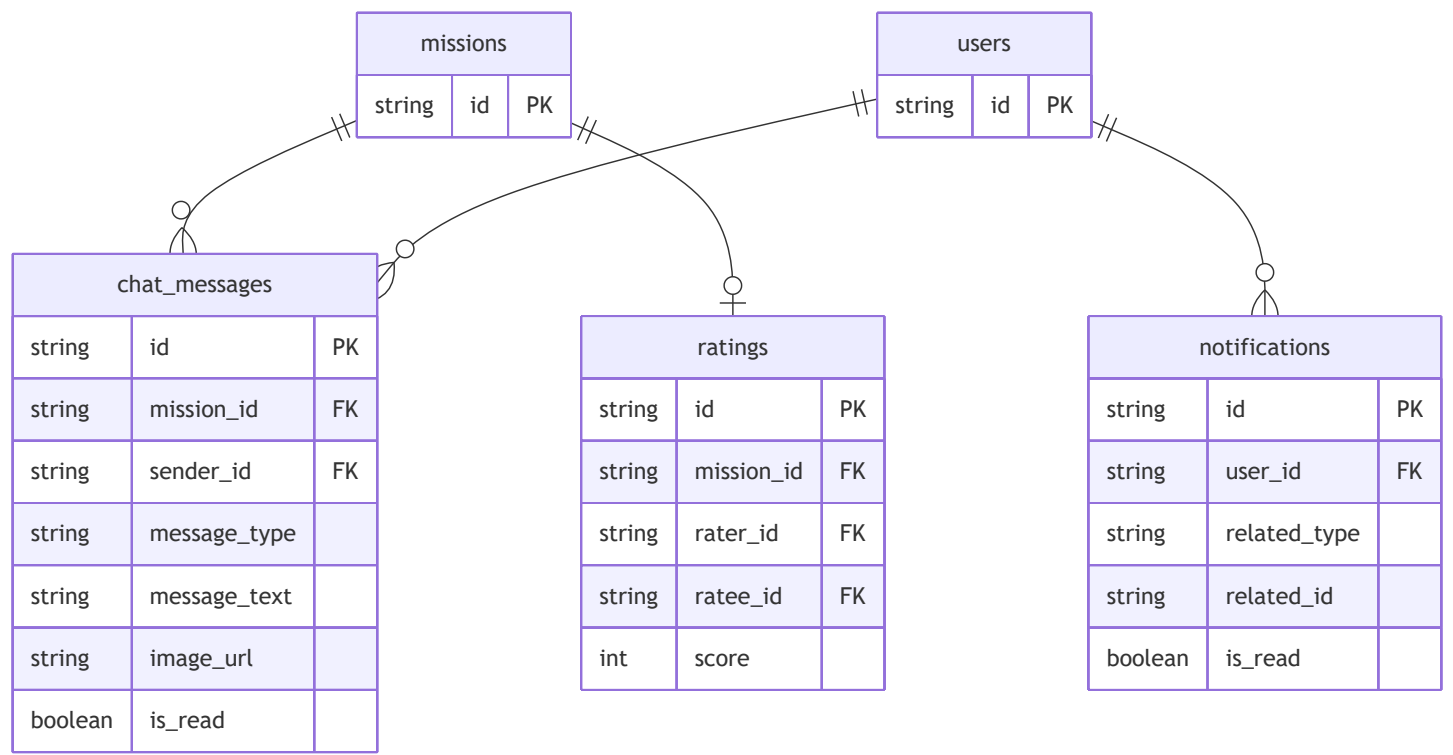
aid_requests		
string	id	PK
string	status	

missions		
string	id	PK
string	mission_type	
string	sos_request_id	FK
string	aid_request_id	FK
string	volunteer_id	FK
string	hub_id	FK
string	status	
string	qr_code_token	
double	priority_score	
string	cancellation_reason	

dispatch_attempts		
string	id	PK
string	mission_id	FK
string	volunteer_id	FK
string	dispatch_type	
int	batch_number	
double	radius_km	
string	response	



E — Communication



4. Cardinality Legend

Ký hiệu Mermaid	Nghĩa
--	Exactly one — Exactly one (1:1 bắt buộc)
--o	Exactly one — Zero or one (1:0-1)
--o{	Exactly one — Zero or many (1:N)
o --o{	Zero or one — Zero or many
}o--o{	Zero or many — Zero or many (M:N)

5. ENUM Reference

ENUM	Giá trị
user_role	VICTIM , VOLUNTEER , SPONSOR , STAFF , ADMIN
hub_status	ACTIVE , INACTIVE , EMERGENCY
urgency_level	CRITICAL , HIGH , MEDIUM , LOW
sos_status	PENDING , DISPATCHING , ASSIGNED , IN_PROGRESS , COMPLETED , CANCELLED
aid_status	PENDING , DISPATCHING , ASSIGNED , PICKED_UP , IN_TRANSIT , COMPLETED , CANCELLED
donation_status	REGISTERED , QR_GENERATED , RECEIVED , REJECTED
mission_type	RESCUE , DELIVERY
mission_status	PENDING , DISPATCHING , ASSIGNED , PICKING_UP , PICKED_UP , IN_TRANSIT , COMPLETED , CANCELLED
dispatch_response	PENDING , ACCEPTED , REJECTED , TIMEOUT
badge_level	BRONZE , SILVER , GOLD , PLATINUM

6. Key Constraints Summary

Bảng	Constraint
users	CHECK (email IS NOT NULL OR phone_number IS NOT NULL)
hub_staff	UNIQUE (hub_id, user_id) WHERE unassigned_at IS NULL
hub_staff	CHECK role = 'STAFF'
hub_inventories	UNIQUE (hub_id, item_category_id) · CHECK (current_quantity >= 0)
inventory_logs	CHECK (quantity_delta > 0)
shelters	CHECK (current_capacity <= max_capacity)

Bảng	Constraint
sos_requests	CHECK (people_count > 0)
aid_requests	CHECK (adults + elderly + children > 0)
missions	CHECK: RESCUE ↔ sos_request_id NOT NULL, aid_request_id NULL, hub_id NULL
missions	CHECK: DELIVERY ↔ aid_request_id NOT NULL, sos_request_id NULL, hub_id NOT NULL
ratings	UNIQUE (mission_id) · CHECK (score BETWEEN 1 AND 5)
chat_messages	CHECK: TEXT XOR IMAGE (không được có cả hai hoặc không có gì)
donation_items · aid_request_items	CHECK (quantity > 0)

7. Changelog

Version	Thay đổi
v3.0	Redesign từ đầu: 22 bảng (bỏ otp_verifications , volunteer_area_experiences , attachments , safe_paths); OTP gộp vào users ; GEOMETRY → lat/lng DECIMAL(9,6); image_url VARCHAR(500) trực tiếp trong bảng; thêm hub_staff.is_available
v2.x	26 bảng — thiết kế cũ (không còn dùng)